

Basic Principles of Medicine 1

Module : Foundations to Medicine

Lecture No: 26

Molecular mechanisms in inherited disorders

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Recommended Reading

- Lippincott's Biochemistry 9th Edition_
 <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lww.healthlibrary.com/book.aspx?bookid=3402>
 - Chapter 3: Globular proteins: Section: Hemoglobinopathies
- Lippincott's Biochemistry 9e, Chapter 3: Hemoglobinopathies; Sickle cell disease; [Globular Proteins | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- Lippincott's Biochemistry: Focused cases: Case 8 – Cystic fibrosis pg: 527; [Clinical Cases | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- Lippincott's Biochemistry Questions: 32.1,32.4

Objectives: Molecular mechanisms in inherited disorders

SOM.MK.I.BPM1.1.FTM.3.BCHM.0179	Describe the molecular basis for clinical features, diagnostic tests and management of Cystic Fibrosis
SOM.MK.I.BPM1.1.FTM.3.BCHM.0180	Describe the molecular basis for clinical features, diagnostic tests and management of Sickle cell anemia
SOM.MK.I.BPM1.1.FTM.3.BCHM.0181	Describe the molecular basis for clinical features and diagnostic tests in Duchenne Muscular Dystrophy and Becker Muscular Dystrophy
SOM.MK.I.BPM1.1.FTM.3.BCHM.0182	Compare the differences between Duchenne muscular dystrophy and Becker muscular dystrophy with reference to severity of the disorder, type of mutations, immunohistochemistry findings and alterations in dystrophin

Molecular mechanisms in inherited disorders

- Most common mutation
- Change in the protein function
- Biochemical basis for the manifestations or symptoms
- Molecular diagnostic tests used and their biochemical basis

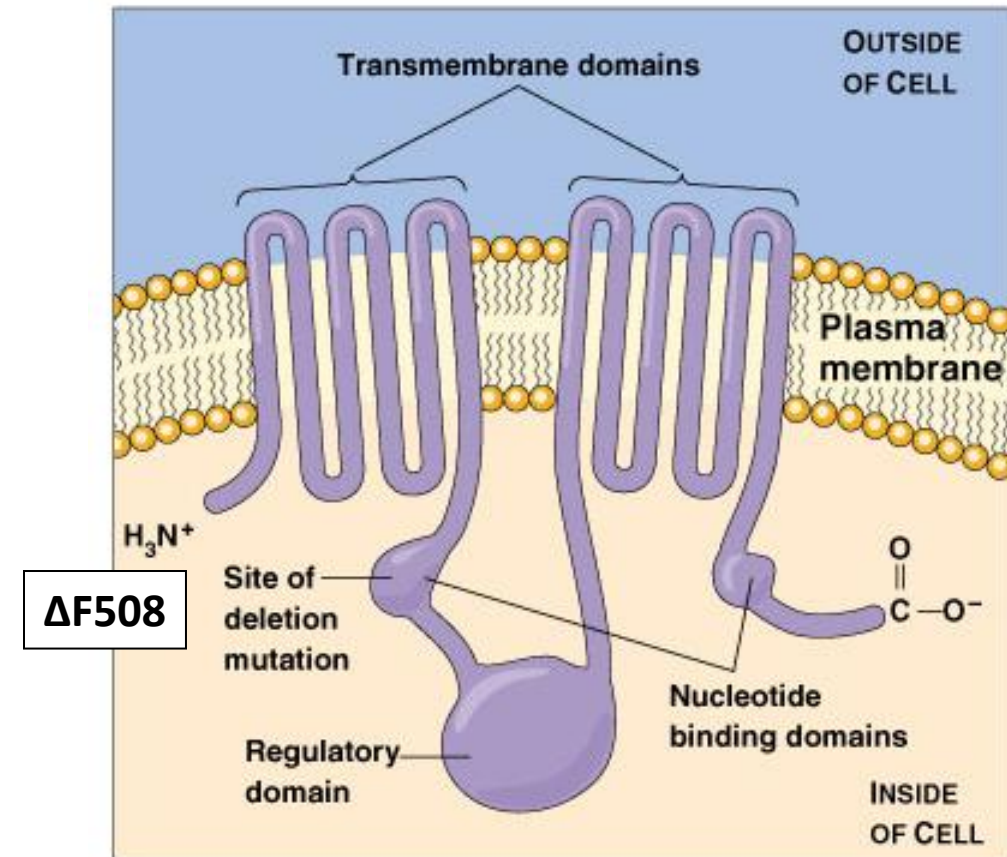
- Cystic fibrosis
- Sickle cell anemia
- Duchenne muscular dystrophy
- Becker muscular dystrophy

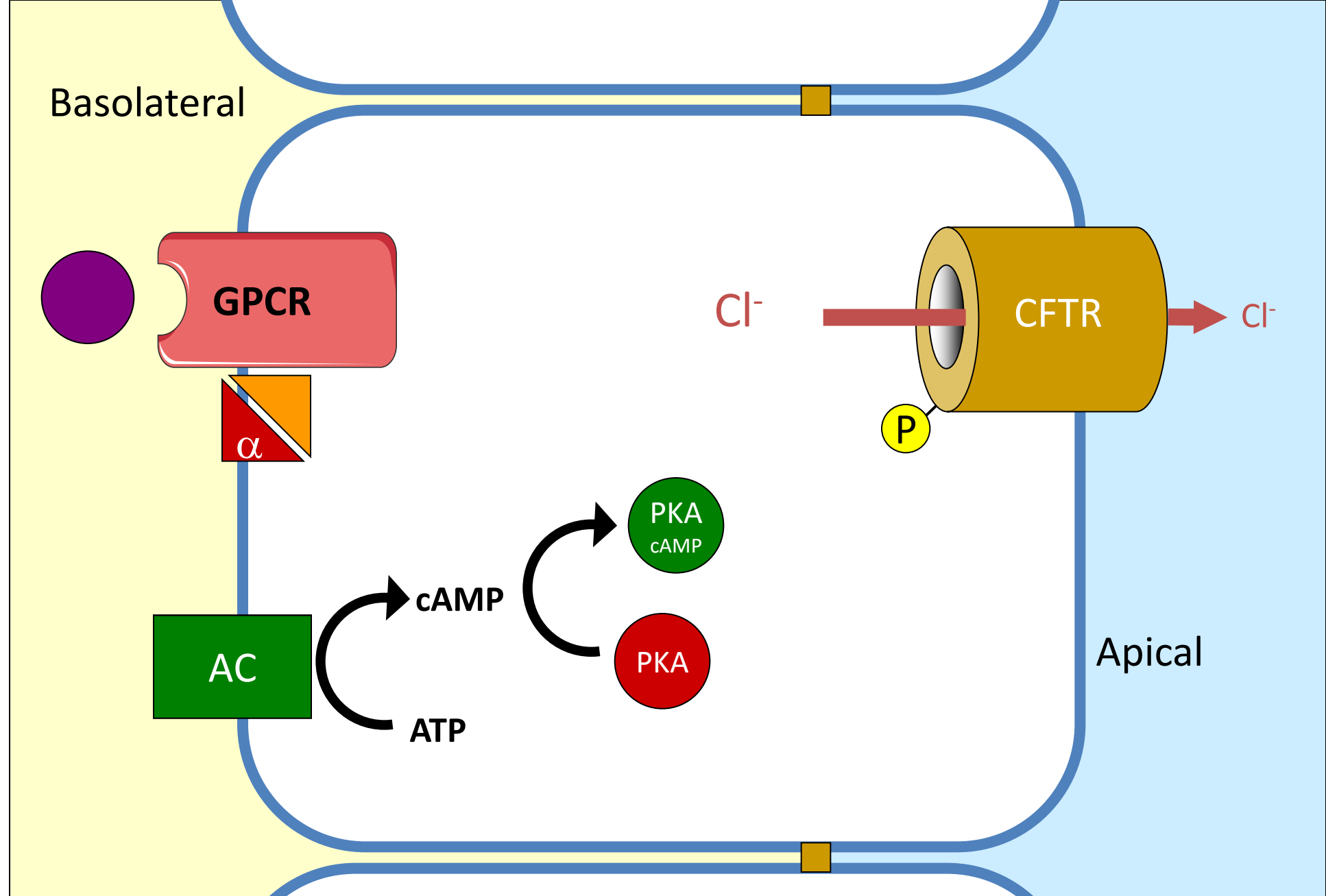
Cystic fibrosis

- Common **autosomal recessive** disorder
- Occurrence: 1 in 2000 in Caucasians
- Mutation in gene for '**Cystic Fibrosis Transmembrane Conductance Regulator**' (CFTR),
- **Chloride channel**
- **ATP Binding Cassette (ABC) transporter**

Structure of CFTR

- CFTR contains
 - Membrane-spanning domains
 - **ATP-binding domains**
 - Regulatory domain that can be phosphorylated by PKA
- Phosphorylation of regulatory domain by PKA activates channel, thus providing **regulated Cl⁻** and fluid secretion





HOW CFTR WORKS AS A REGULATED Cl^- CHANNEL

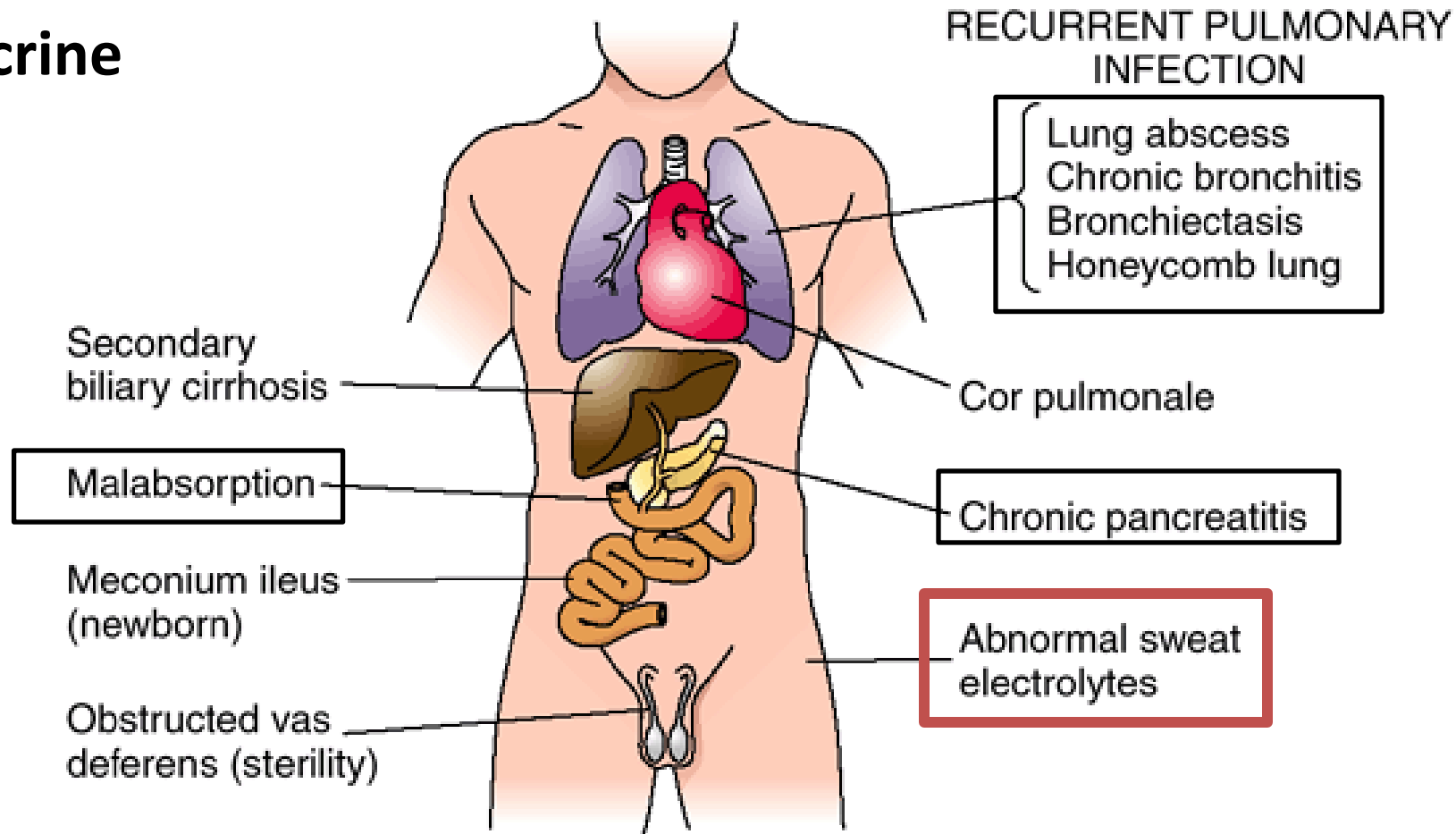
Dr. Dasso

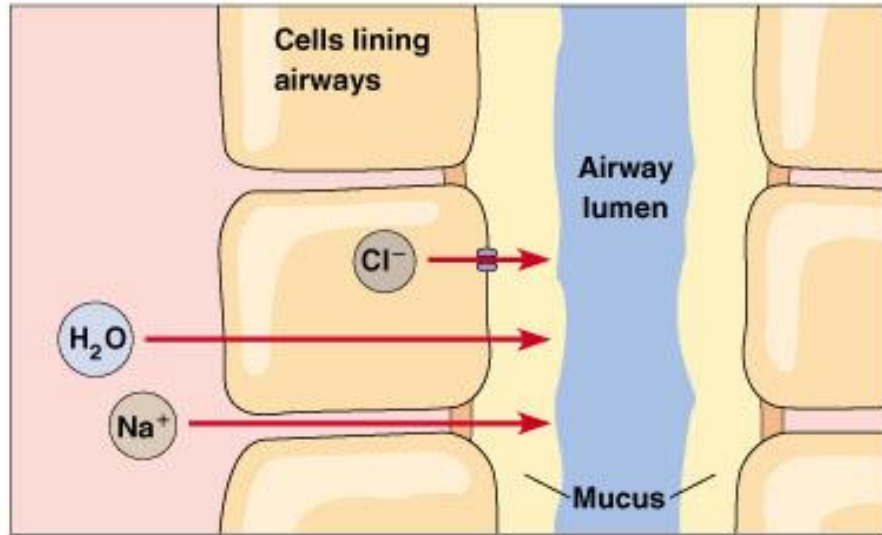
Case report on cystic fibrosis (Mark's textbook)

- Sissy Fibrosa, has had repeated bouts of **bronchitis (respiratory infections)**. She also has **foul smelling, bulky stools (steatorrhea)**
- Growth records show a **decline** in rate of **growth**
- Low level of serum proteins and serum albumin (**chronic malnutrition**)

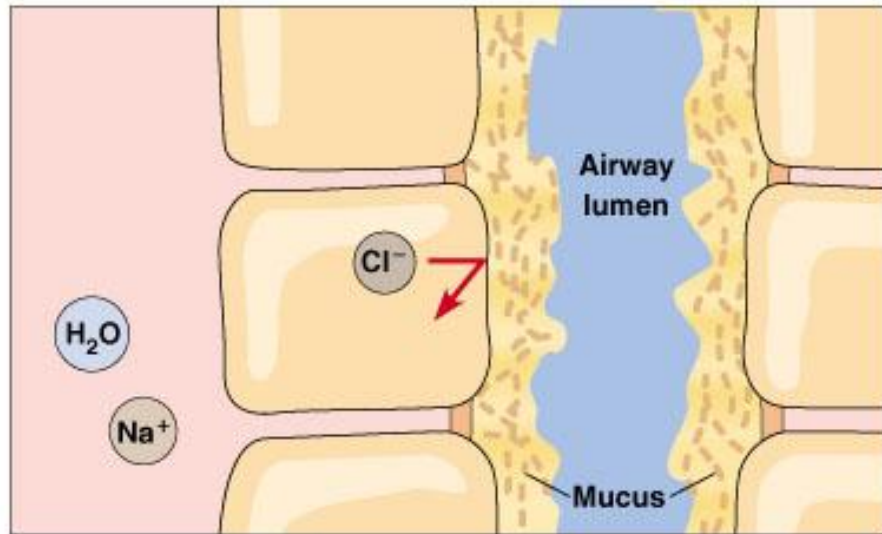
Clinical features of cystic fibrosis

Affects exocrine glands





Normal cells lining airways; hydrated mucus



Cells of a person with cystic fibrosis;
dehydrated mucus infected with bacteria

(a) Normal and faulty chloride ion secretion

Respiratory system in cystic fibrosis

- Mutant CFTR does **not** transport Cl^- into airway lumen
- As a result, Low Na^+ and H_2O content of luminal secretions → **Thickened and viscid mucus secretions**
- Viscid secretions prone for **bacterial infections**
- Respiratory infections: **Mortality and morbidity**

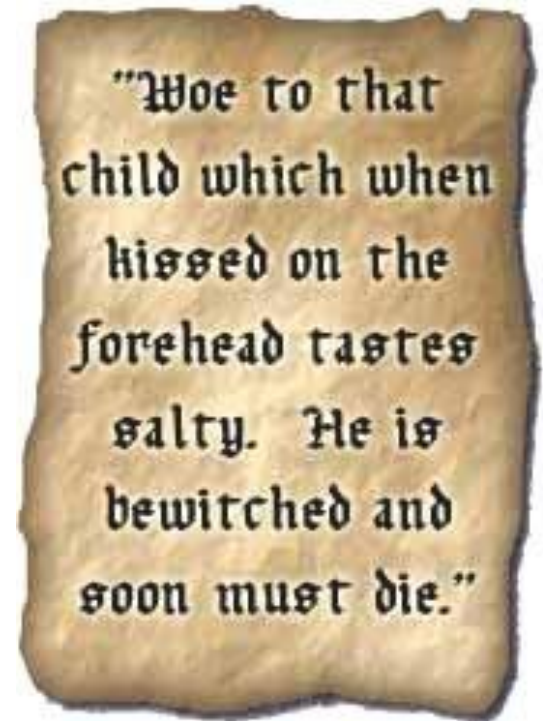
Changes in the pancreas in cystic fibrosis

- **Exocrine pancreatic insufficiency**
- Loss of CFTR leads to **thicker** acinar secretions within duct lumen: **duct obstruction and tissue destruction**.
- Fibrotic tissue and fat replace pancreatic parenchyma - “**cystic fibrosis**” → Deficiency of pancreatic enzymes
- Nutrient maldigestion and loss of fat in stools (**steatorrhea**)
 - Deficiency of fat-soluble vitamins
- **Protein malnutrition and growth delay**
- Oral enzyme replacement therapy improves nutrition
- **Deficient secretion of pancreatic enzymes** (lipase, trypsin, chymotrypsin)
 - Normal digestion and nutrition restored by pancreatic enzyme supplements



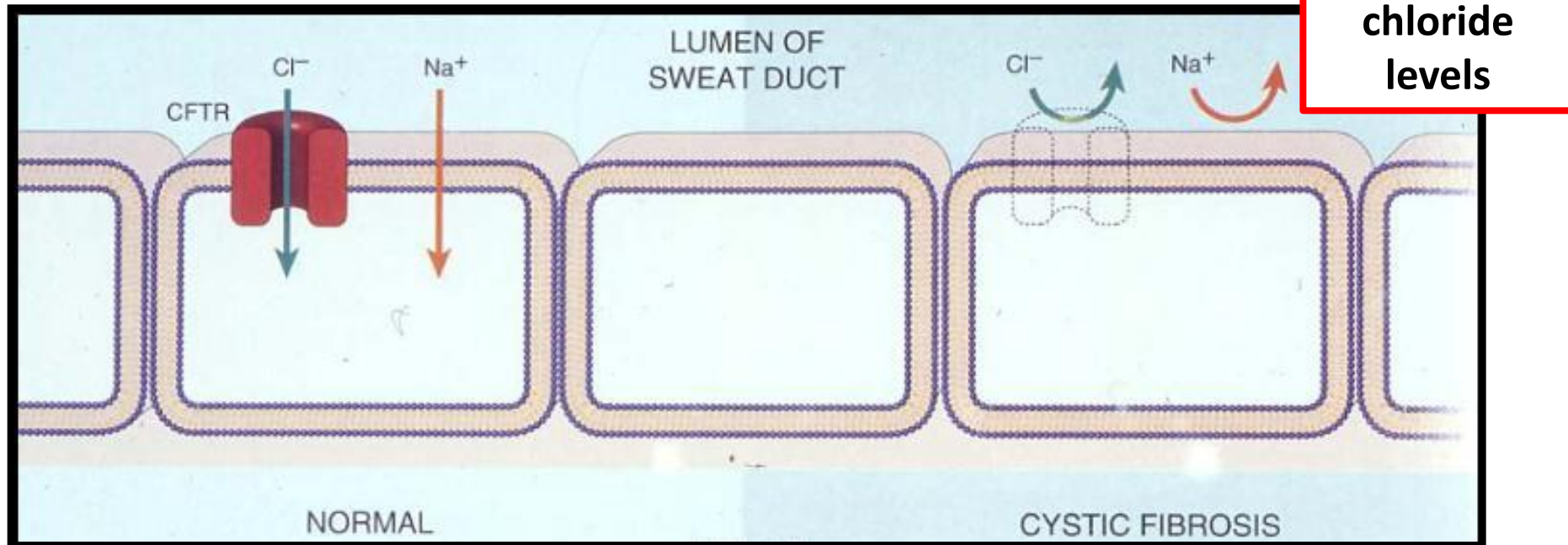
Other systems involved in cystic fibrosis

- **Meconium ileus and intestinal obstruction** neonates
 - Due to viscid intestinal secretions
- Infertility in males - lack vas deferens, known as **congenital bilateral absence of vas deferens (CBAVD)**
- <https://www.youtube.com/watch?v=qmzvCnt5cHA> (Day in the life of a cystic fibrosis patient)



18th century German saying

Changes in the sweat glands



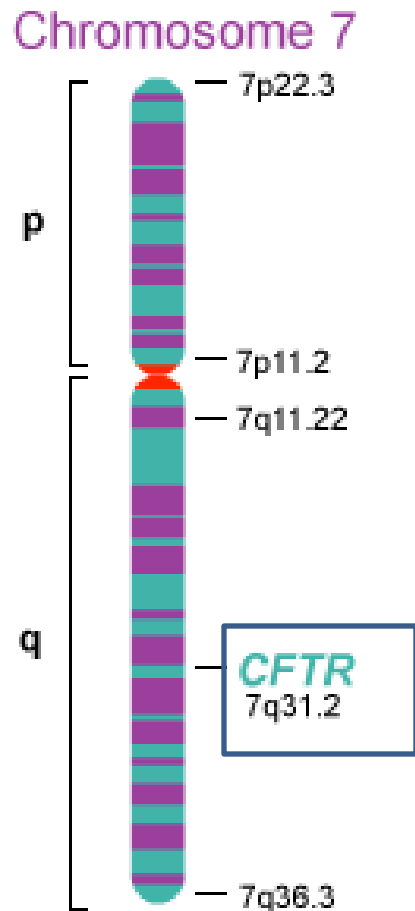
- In sweat glands, CFTR involved in **reabsorption** of NaCl.
- In CF very little NaCl is reabsorbed, resulting in **high sweat salt content**.
- **High sweat chloride** levels - diagnostic test for CF

DIAGNOSIS OF CF: SWEAT CHLORIDE TEST

- Pilocarpine placed under electrode pad
- Mild electric current passed between electrodes to force pilocarpine into skin, to stimulate sweat glands.
- Sweat collected and chloride levels measured
- **Elevated sweat chloride** levels are diagnostic

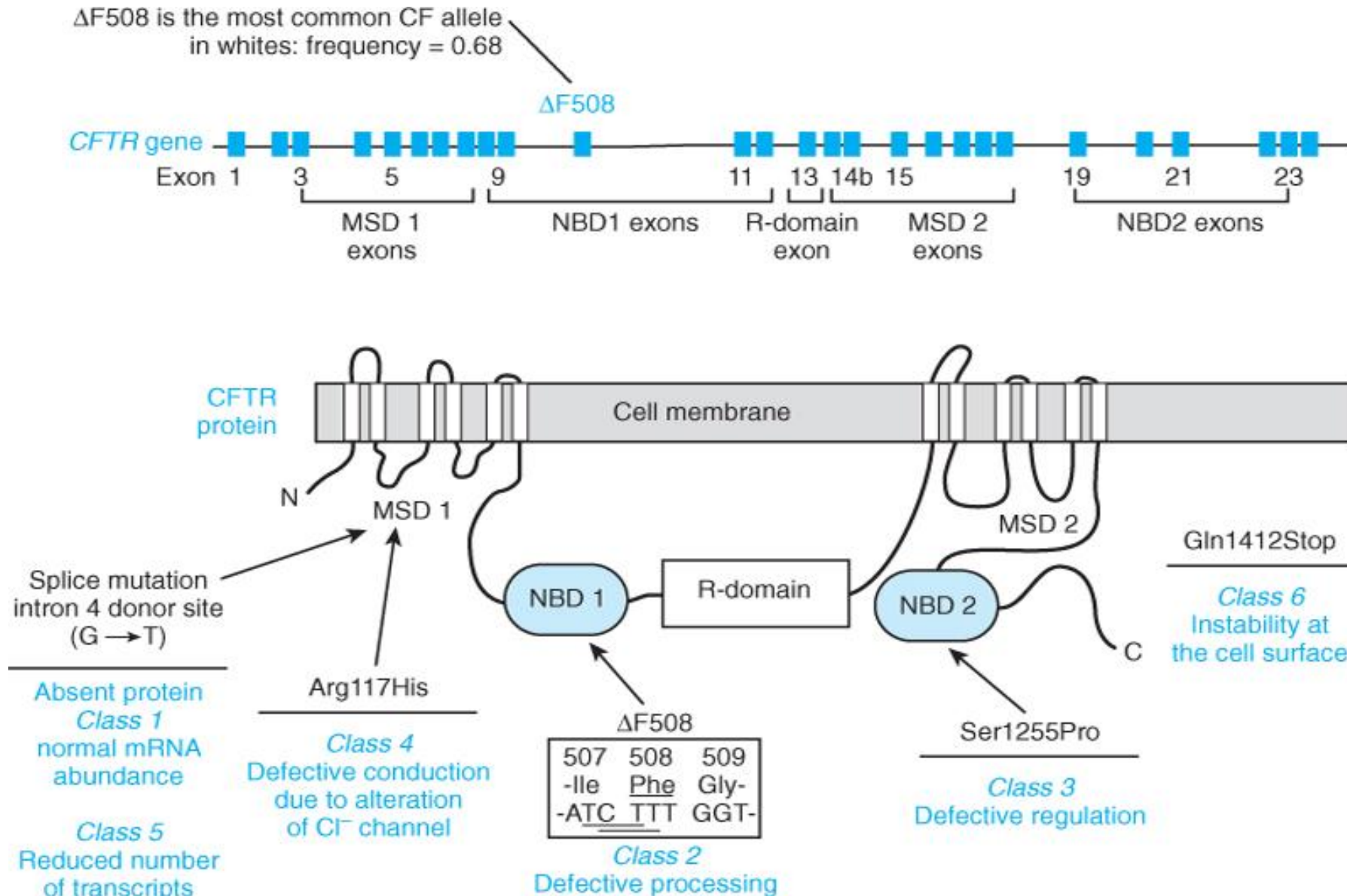


CFTR mutations: Genotype-phenotype correlations



- Autosomal recessive single gene disorder
- CFTR locus on **long arm of chromosome 7**
- >1000 cystic fibrosis mutations
- Most common (about 70%) is $\Delta F508$
 - 3-bp deletion
 - Eliminates **phenylalanine** at position 508: $\Delta F508$ mutation
- $\Delta F508$ mutation prevents maturation of protein and reaching plasma membrane (**severe phenotype**)
- Other CFTR mutations associated with variable severity

CFTR mutations: Genotype-phenotype correlations



CFTR mutations characterized by **allelic heterogeneity**

Molecular diagnostic tests

- **ASO test**: Useful if the mutation is known
- Carriers are **heterozygous** for the mutation, whereas, affected children are **homozygotes** for the mutation
- Remember, due to **allelic heterogeneity** patients may be **compound heterozygotes**

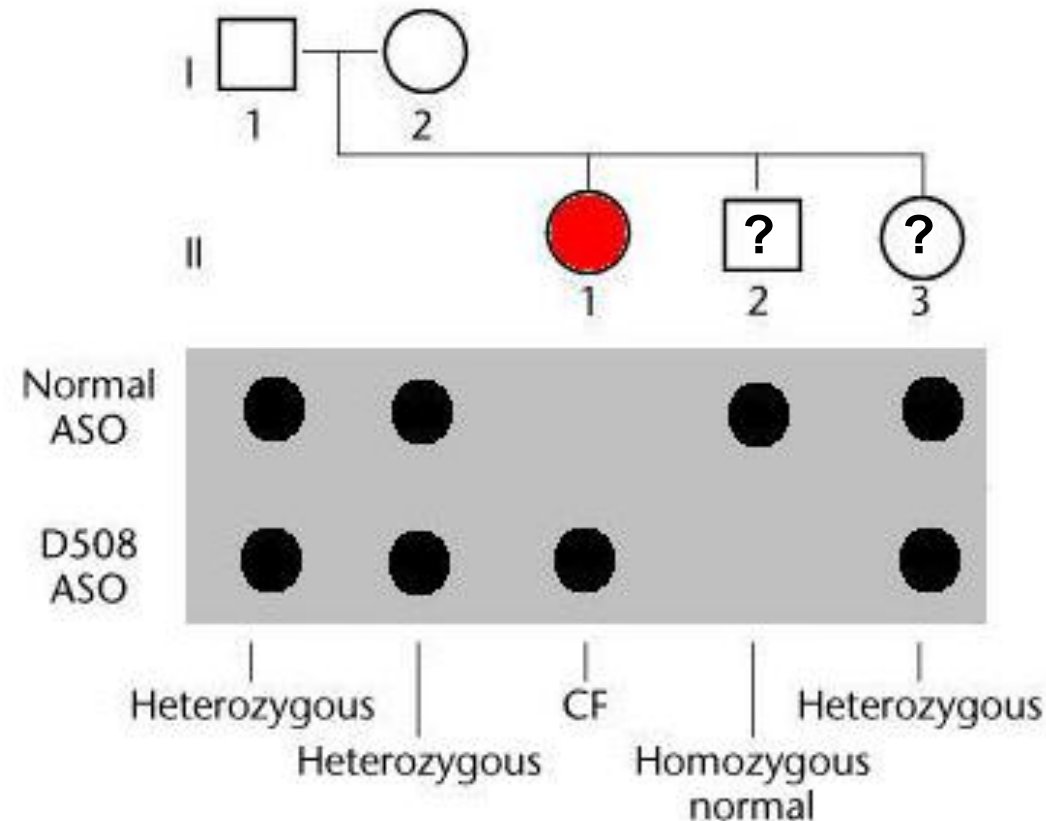
Dot Blot Assay of cystic fibrosis using ASO probes

Cystic Fibrosis allele $\Delta 508$ has 3bp deletion [AGA]

ASO for normal DNA 5' CACCAA $\boxed{\text{AGA}}$ TGATATTTTC-3'

ASO for DNA sequence of $\Delta 508$ mutation 5' CACCAATGATATTTTC-3'

The most common mutation
in cystic fibrosis is a
deletion of codon 508 (Phe)



**Carrier of the
mutant gene**

Sickle cell anemia (Mark's textbook)

- Will Sichel, a 17-year-old boy presents to ER with severe lower back and abdominal pain
- Diagnosed as having sickle cell disease at 3 years of age
- Lab results on admission
 - Hemoglobin = 6.8 gm/dL (Normal: 12-16gm/dL)
 - Total bilirubin = 2.3 mg/dL (Jaundice)
- Vaso-occlusive sickle cell crisis (hemolytic crisis)
- Autosomal recessive disorder
- <https://www.youtube.com/watch?v=2CsgXHdWqVs> **Sickle Cell Anemia: A Patient's Journey;** <https://jamanetwork.com/journals/jama/article-abstract/375924>: First patient from Grenada

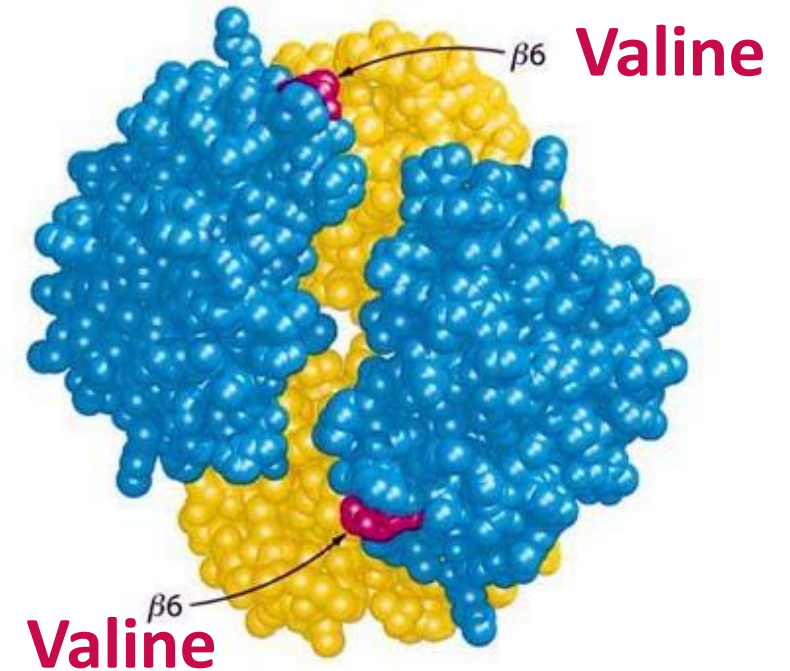
Molecular defect in Sickle cell anemia

	Thr	Pro	Glu	Glu	beta ^A chain
	... A C T	C C T	G A G	G A G ...	beta ^A gene
Codon #	4	5	6	7	
	... A C T	C C T	G T G	G A G ...	beta ^S gene
	Thr	Pro	Val	Glu	beta ^S chain

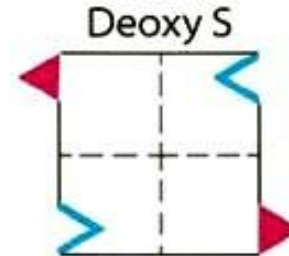
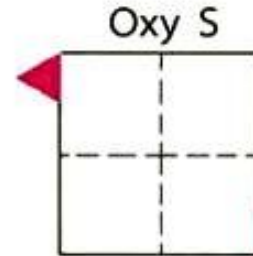
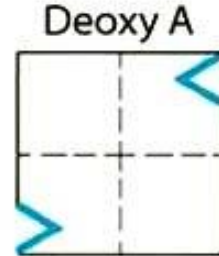
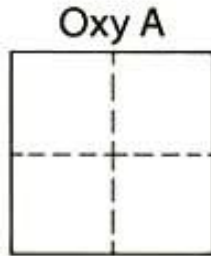
- Point mutation in **β-globin** gene of hemoglobin
- Glutamic acid replaced by **valine** at sixth position of β-globin chain
- Acidic amino acid (hydrophilic) replaced by **branched chain (hydrophobic)** amino acid (**MISSENSE**)
- Glutamic acid present on exterior of hemoglobin

Molecular change in sickle cell disease

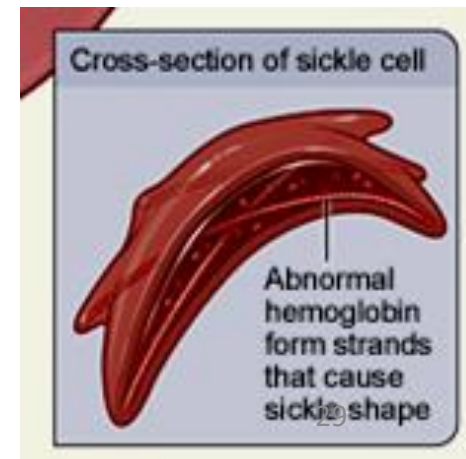
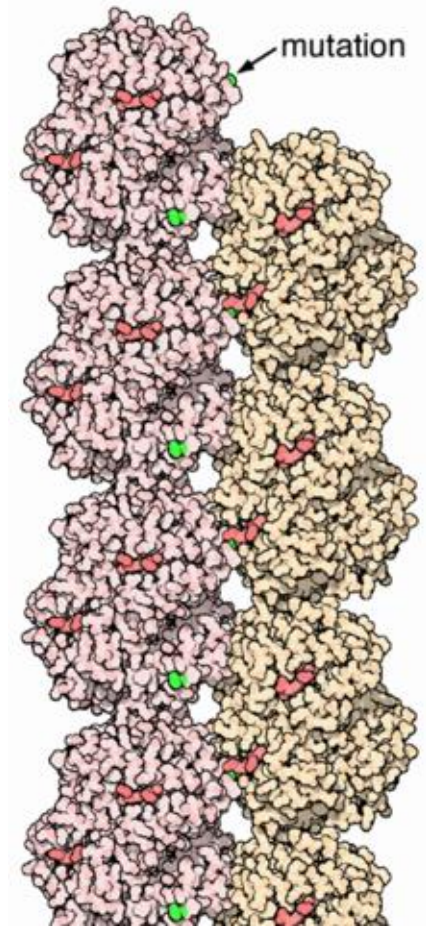
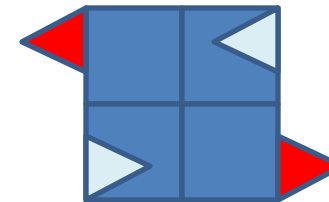
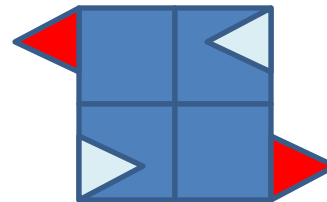
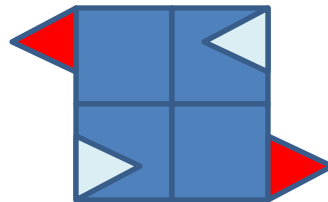
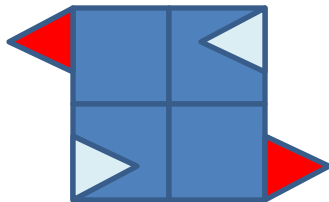
- Replacement by **valine** in beta-globin chain creates a **hydrophobic pocket** on exterior of hemoglobin molecule
- In deoxygenated state, HbS **aggregates** to form long filaments
- HbS Aggregation results in RBC distortion – sickling
- Hydroxyurea in management



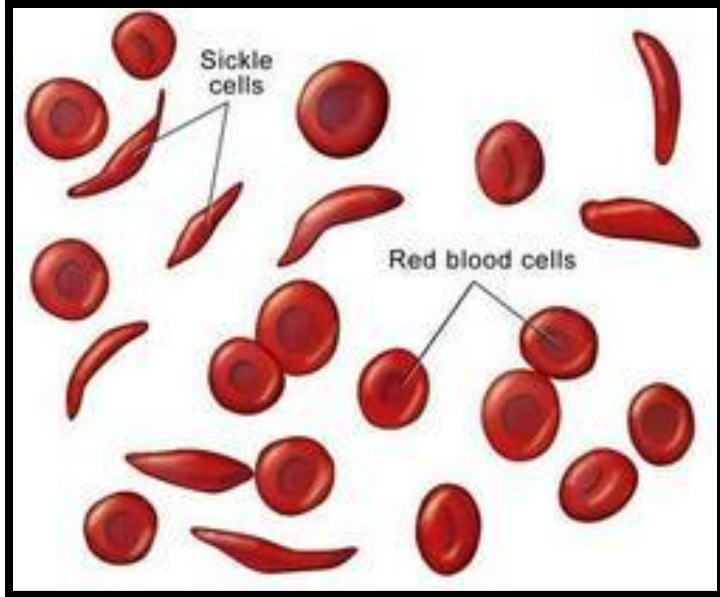
Changes in deoxy HbS



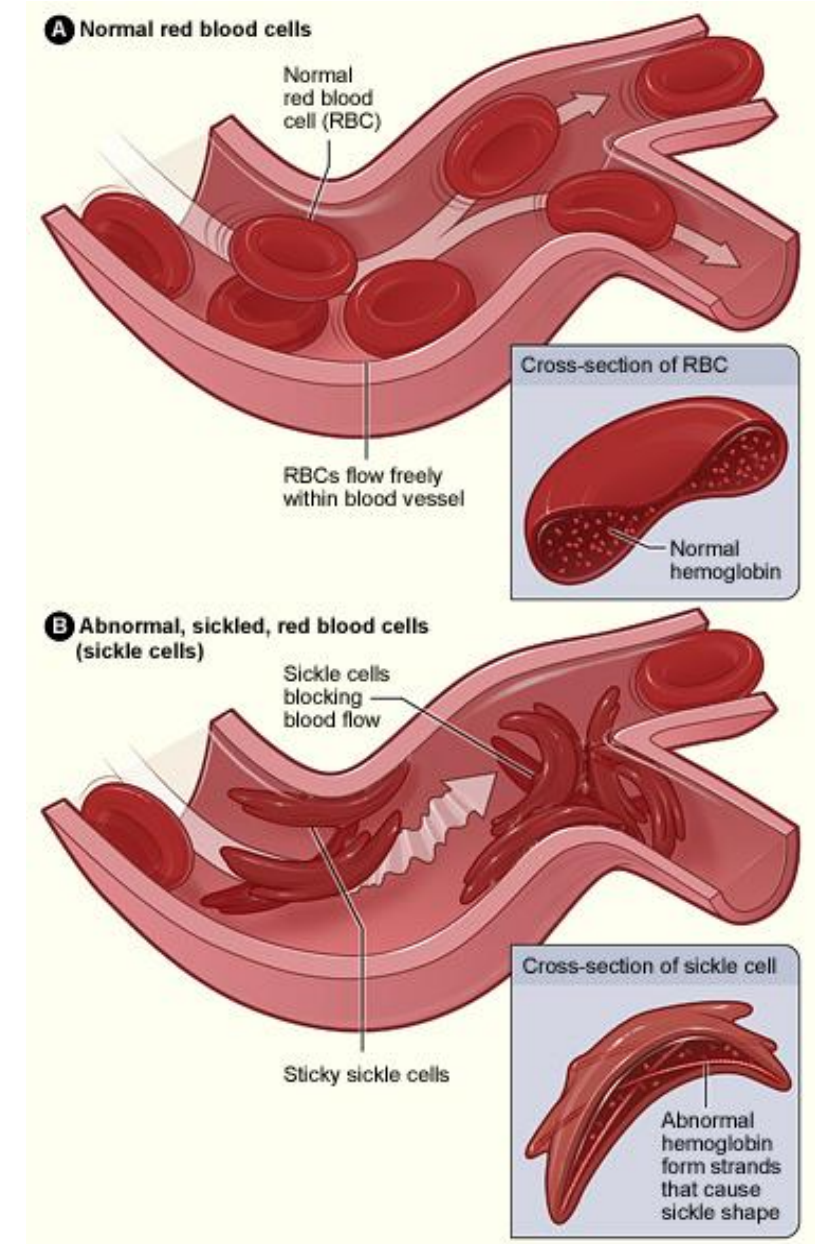
Hydrophobic 'sticky' patch in HbS (Glu→Val)



Changes during sickling crisis



Peripheral blood smear showing sickled red cells



Clinical features in Sickle cell disease

- Sickled and distorted RBCs are removed by spleen
- Hemolytic anemia (Less lifespan of RBC)
- Splenomegaly
- Sickling (Hemolytic) crisis results in excessive removal of sickled RBC, resulting in worsening of anemia and jaundice (hyperbilirubinemia)
- Silent strokes

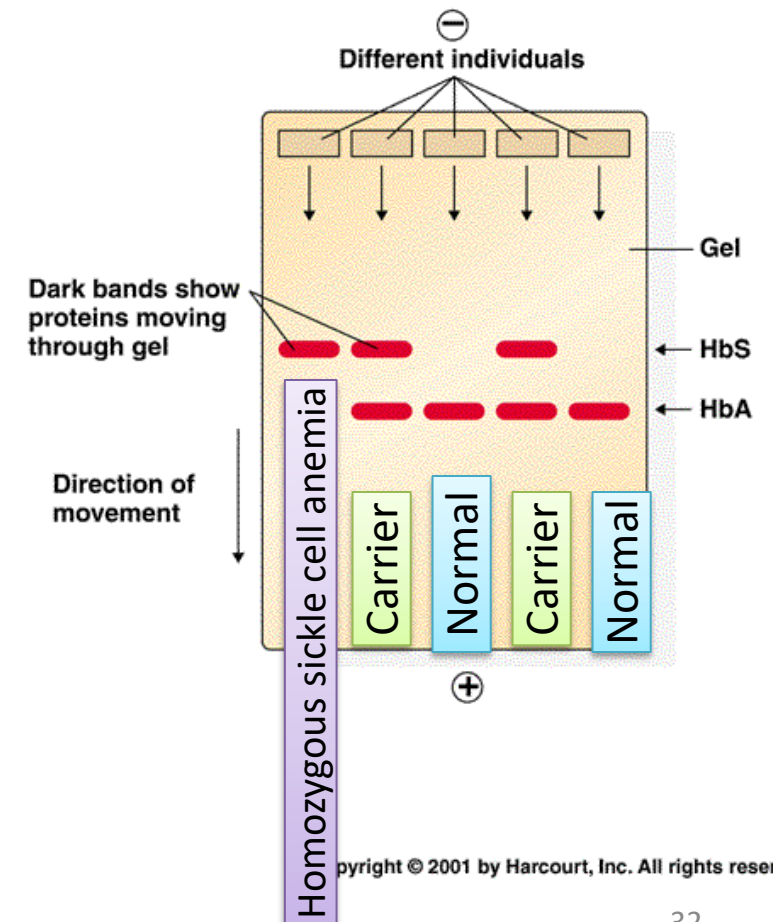


Jaundice

Hemoglobin electrophoresis

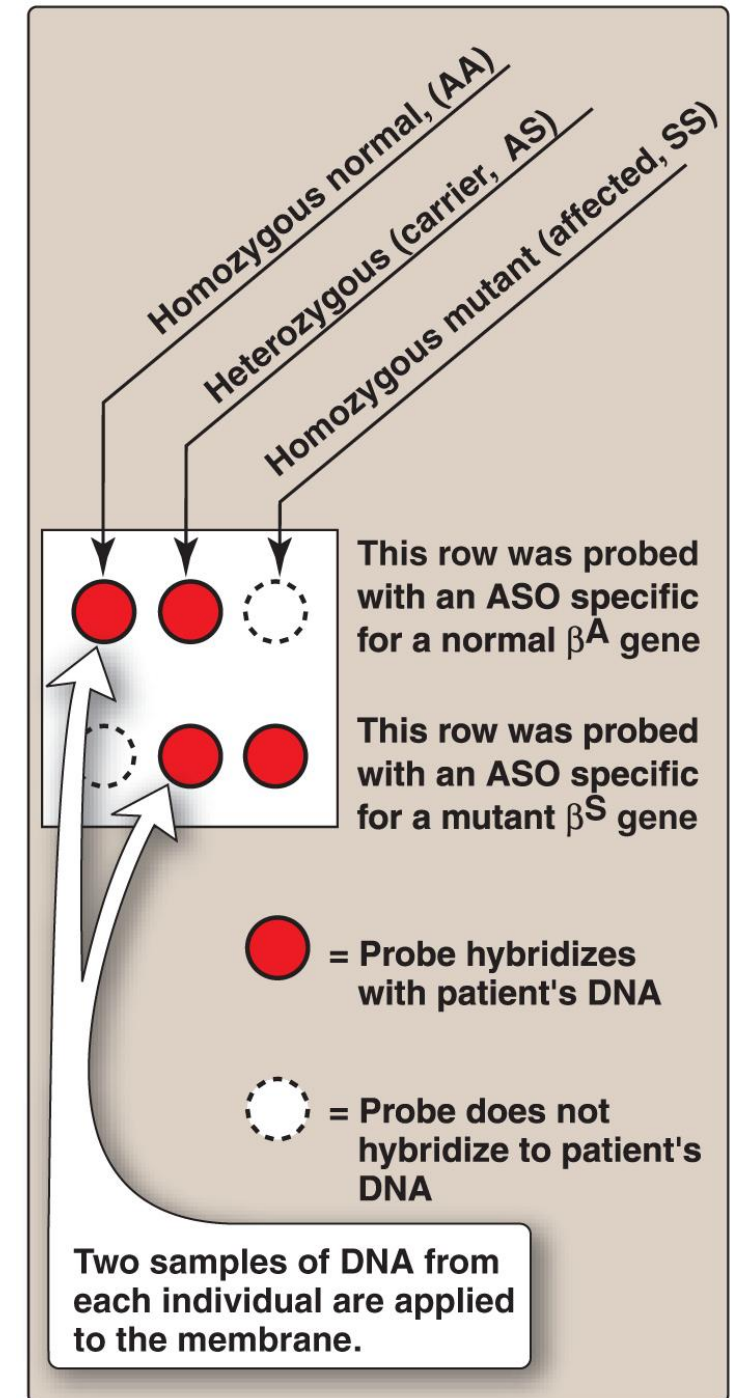
Tobin/Dusheck, Asking About Life, 2/e
Figure 10.15

- Hemoglobin electrophoresis: HbS (**valine**) has **fewer negative charges** than HbA (glutamate)
- HbS moves slower than HbA towards anode (+)



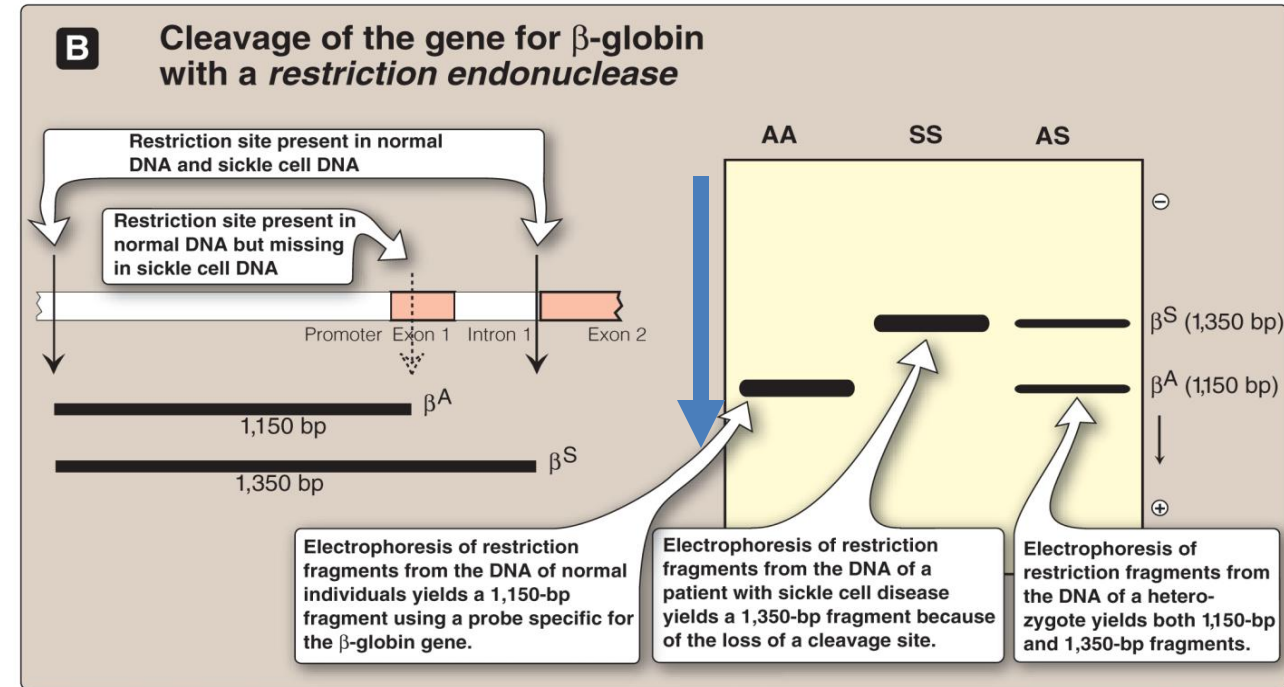
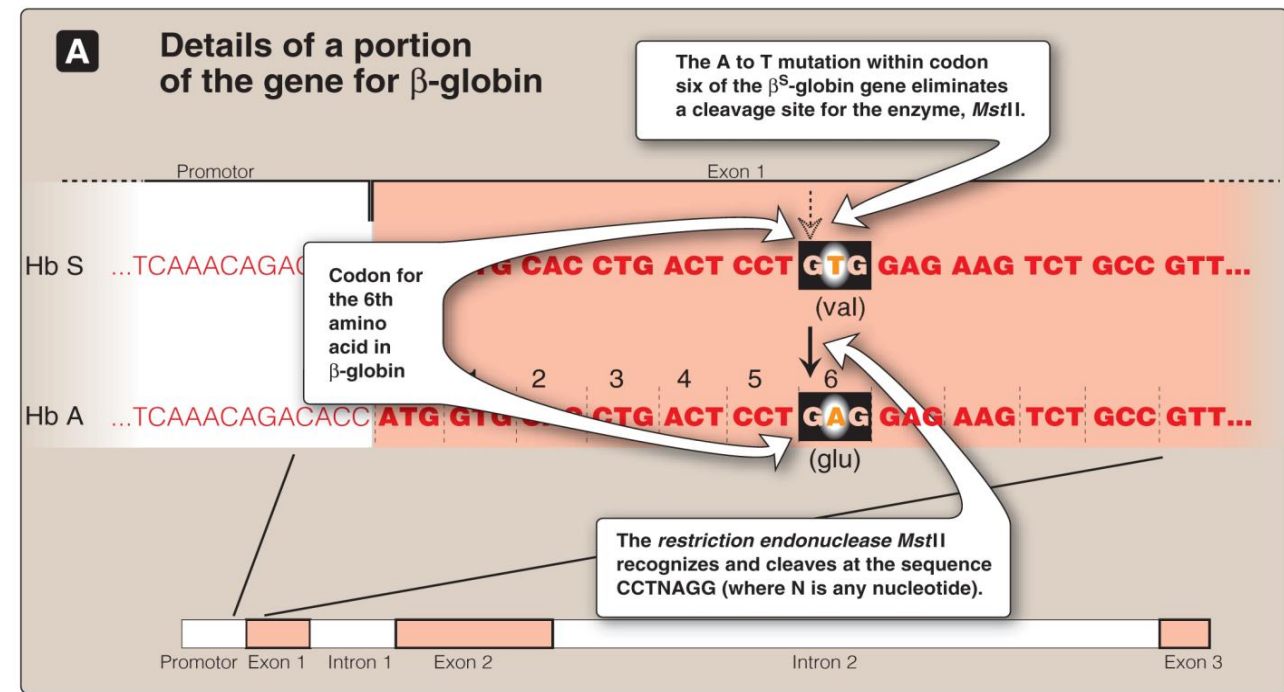
ASO test for HbS

- ASO test (Dot blot test) using a specific probe for HbS
- **Heterozygous carriers** (one copy of normal allele and one copy of mutant allele) and **homozygous affected**



RFLP test for detection of sickle cell disease and carriers

- Mutation results in loss of restriction site on β -globin gene –results in larger fragment being generated
- Size of fragment from HbS is larger (moves slower; 1350 bp) compared to fragment from HbA (moves faster; 1150bp)



Duchenne Muscular Dystrophy And Becker Muscular Dystrophy

- Both disorders are due to mutations in **dystrophin** gene
- Largest gene located on **X-chromosome**
- Both are **X-linked recessive** disorders
- DMD is due to almost complete **absence** of functional dystrophin
- BMD is due to production of **abnormal dystrophin or less amounts of dystrophin**

Duchenne Muscular Dystrophy

- Absence of **dystrophin**, muscle protein
- X-linked recessive disorder; **Males**
- Most common muscular dystrophy, affecting 1 in 3,000 boys.
- Age of onset: 2-6 years
- Death in early 20s due to breathing and heart problems; Wheel-chair bound in their teens.
- [Gowers sign](#)
- Males with DMD have **low reproductive fitness (unable to pass the mutation to the next generation)**

Becker Muscular Dystrophy

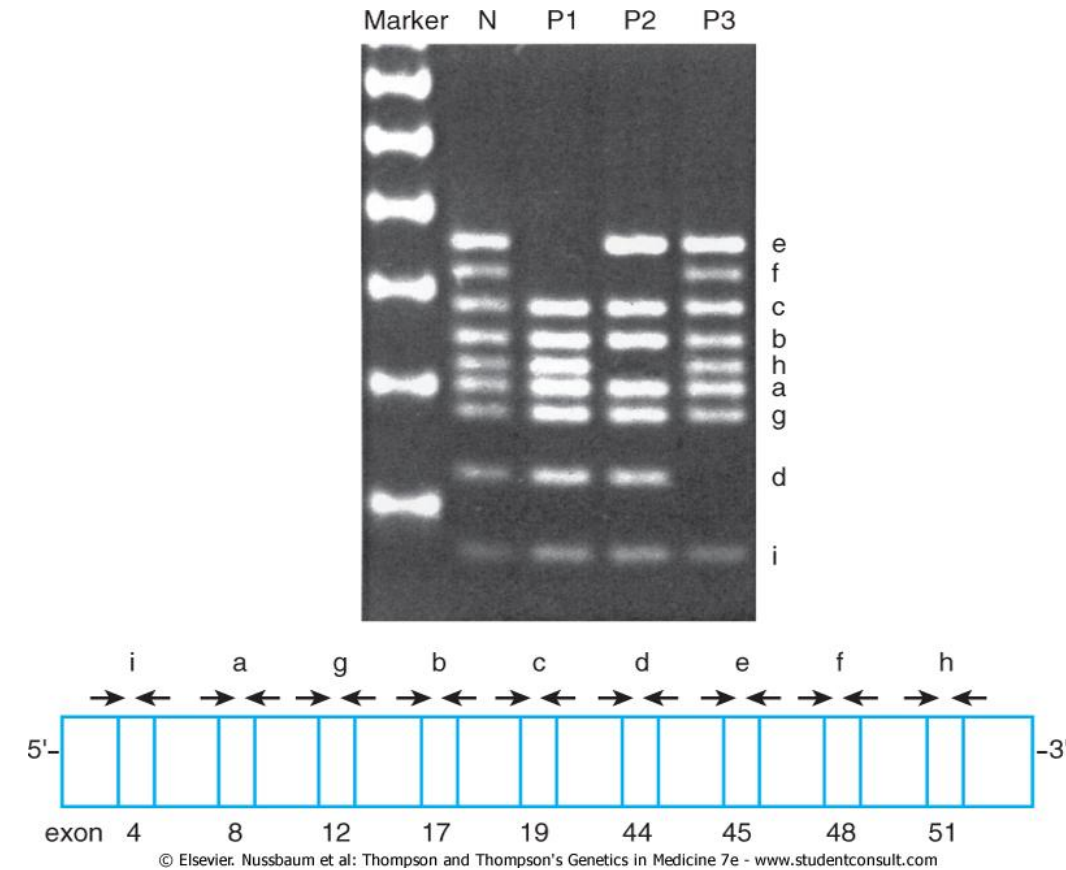
- Some dystrophin present; truncated dystrophin
- Milder form of muscular dystrophy
- X-linked recessive; Males
- Symptoms similar to DMD but milder
- Age of onset: 20-30 years; Milder course of disease
- BMD affects 1 in 30,000 live male births.
- Males with BMD typically have **normal reproductive fitness**

Mutations in DMD and BMD

- Both DMD and BMD result from mutation in **dystrophin** gene
- DMD deletions can be distinguished from deletions causing BMD
 - Deletions causing DMD cause **frameshift mutations**
 - Mutations also involve **large deletion of exons**
 - **NO Dystrophin synthesis**
 - Deletions in BMD are **in-frame deletions or promoter mutations**
 - **Some dystrophin**, although truncated, is translated
 - Or reduced synthesis of dystrophin

DMD: Multiplex PCR diagnostics

Multiplex PCR: Simultaneous amplification of many DNA segments in one amplification reaction



Diagnosis of DMD involves screening for deletions and duplications by a procedure called multiplex polymerase chain reaction (PCR). Lane 2 shows PCR products from normal (N), indicating presence of corresponding exons. Patient 1 (lane P1) is missing bands e and f, a deletion encompassing exons 45 to 48. Patient 2 (lane P2) is missing bands f and h, deletion involving exons 48 to 51. Patient 3 (lane P3) is missing band d and has deletion of exon 44.

Clinical features of DMD

Pseudohypertrophy of calf



© Elsevier. Nussbaum et al: Thompson and Thompson's Genetics in Medicine 7e - www.studentconsult.com

Pseudohypertrophy of calves due to replacement of muscle tissue with connective tissue and fat in an 8-year-old boy with DMD

Early stages, Duchenne and Becker MD affect pectoral muscles, trunk, and upper and lower legs. Difficulty in rising, climbing stairs and maintaining balance.



Because of weakened leg muscles, boys with DMD have a distinctive way of rising from the floor, called **Gowers' maneuver**. First get on hands (1) and knees (2), then elevate the posterior (3), then "walk" their hands up the legs (4) to raise their upper body.



1



2



3

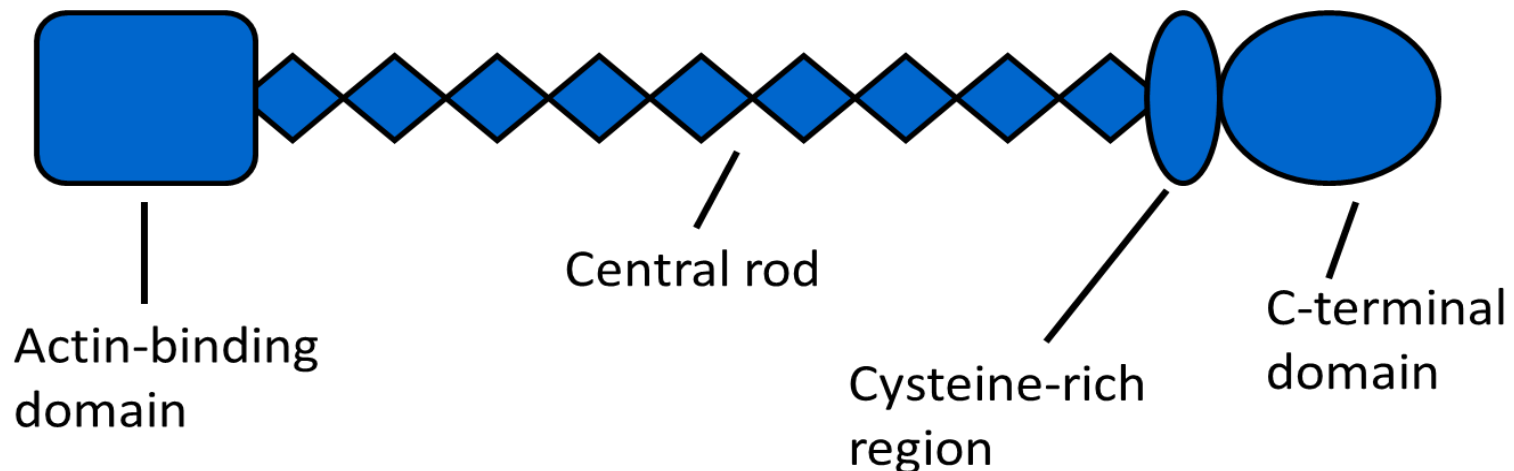


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[Gowers sign](#)

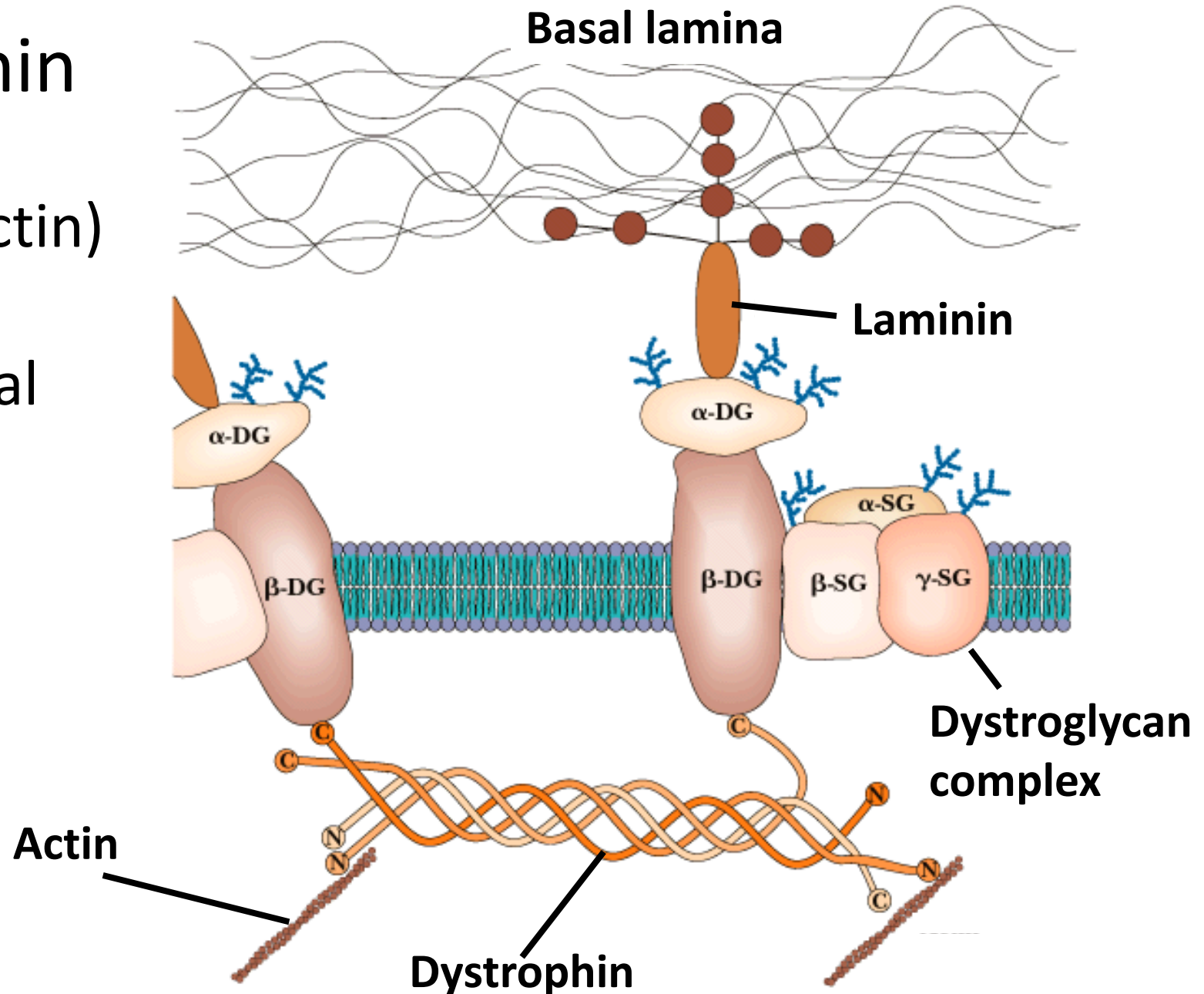
Dystrophin gene and protein

- **Largest gene** Almost 2% of X-chromosome and nearly 0.05% of genome. 79 exons and 8 promoters; 2.2 million base pairs.
- **Mutation hotspots** – almost 33% of patients have new mutations
- Expressed in **smooth, cardiac, and skeletal muscle**, with lower levels in the brain.



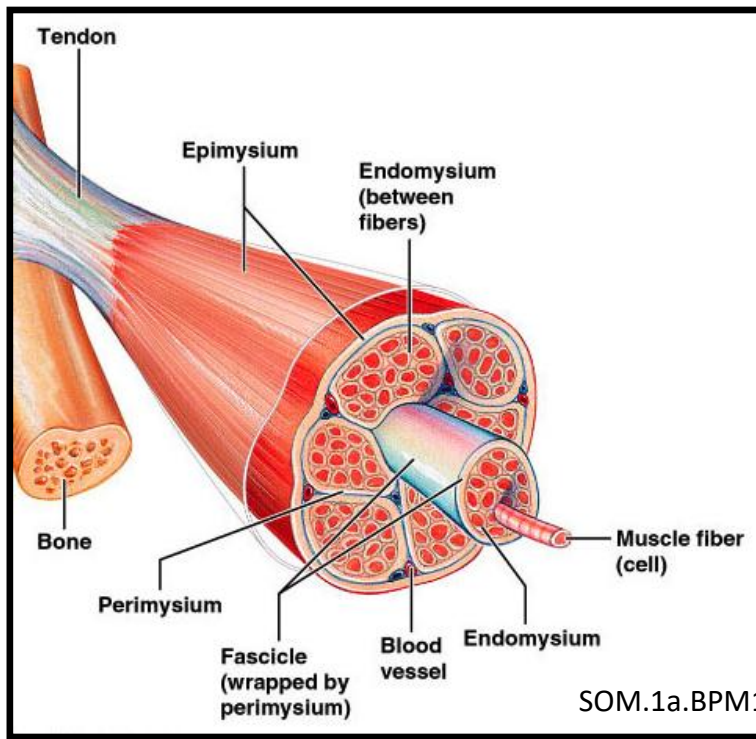
Function of dystrophin

- Anchors cytoskeleton (actin) of muscle cells to extracellular matrix (basal lamina)
- Links actin filaments to proteins of muscle cell membrane

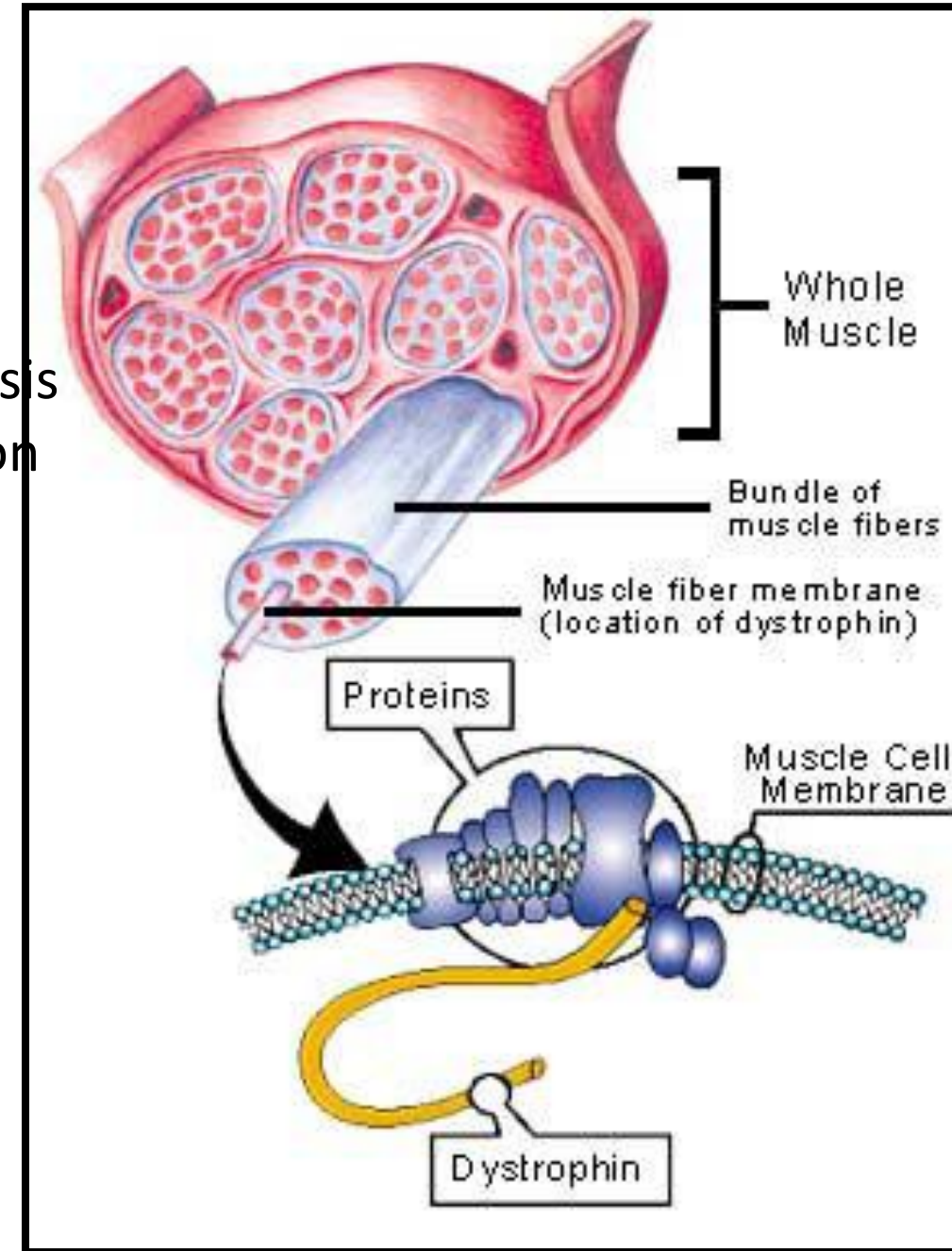


Dystrophin protein

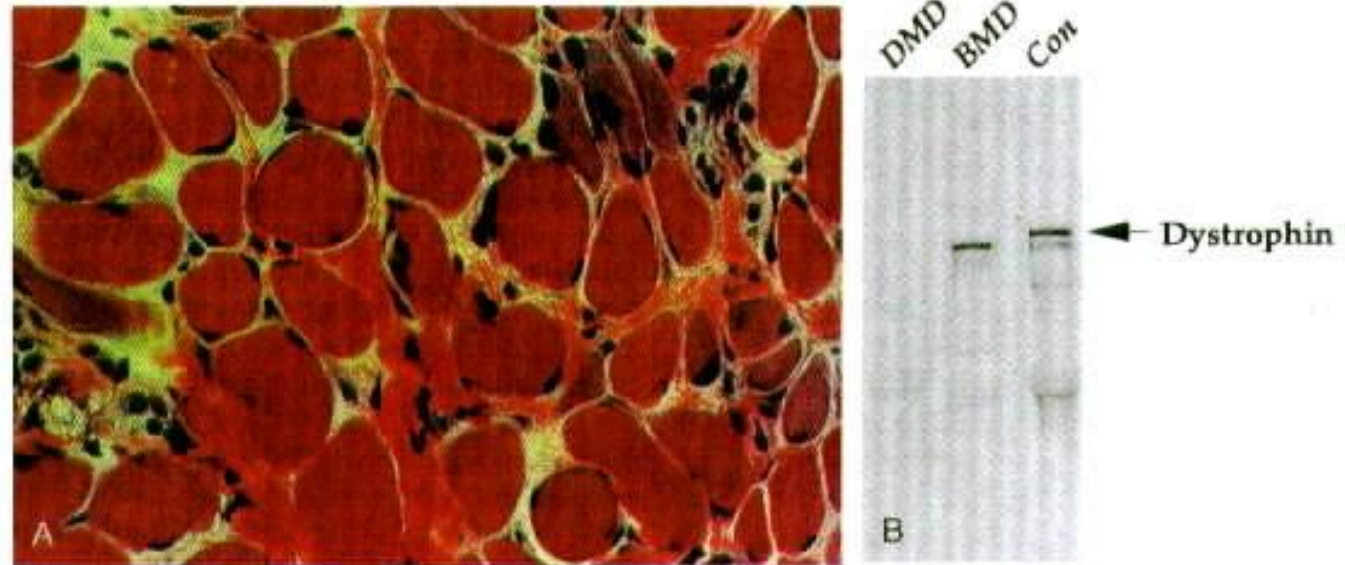
Enables muscle to withstand stress of muscle contraction
Loss of dystrophin leads to oxidative injury and myonecrosis
Muscles susceptible to mechanical injury and degeneration



SOM.1a.BPM1.1.FTM.3.BCHM.0181,0182



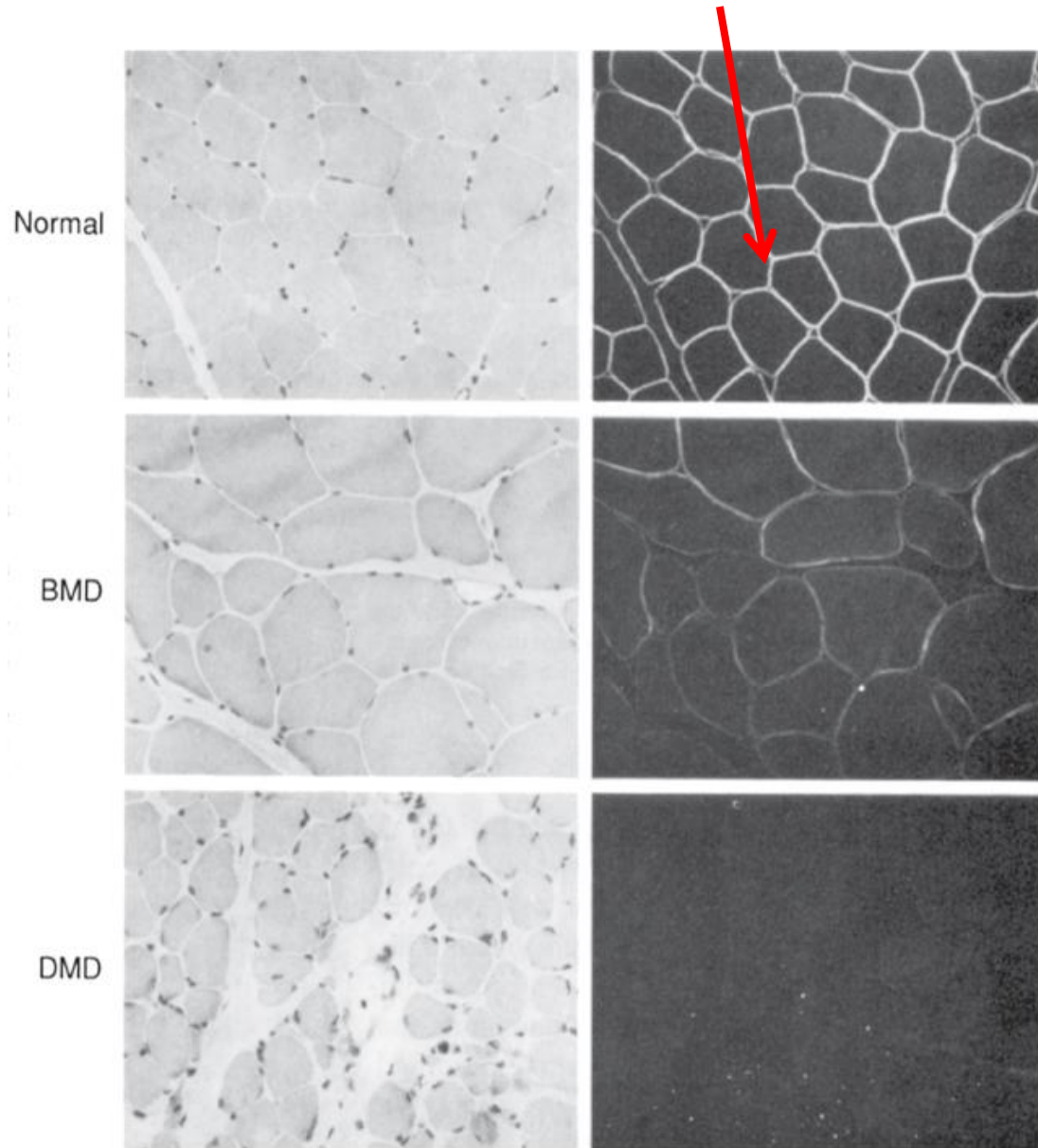
DMD Vs BMD: Histology and Western blot



- Histological staining shows an increase in connective tissue between myocytes
- **Western blot**
 - Smaller protein size, and may be reduced quantity of dystrophin protein in BMD (**In-frame deletion**)
 - Complete absence of dystrophin in DMD (**Frame-shift deletion**)

Dystrophin antibody
staining of muscle cells

DMD Vs BMD: Immunostaining



- Microscopy in BMD and DMD
- Left: Muscle Hematoxylin and eosin staining
- Right: Immunofluorescence microscopy staining with dystrophin specific antibody
- **Reduced quantity of dystrophin in BMD, and complete absence of dystrophin in DMD**
- Increased Connective tissue between myocytes in DMD muscle

Other tests useful in Duchenne muscular dystrophy

- Serum **creatine kinase (MM) levels** elevated in patients with muscular dystrophy (indicative of muscle damage)
- Females are carriers and they usually have higher levels of **CK-MM** in serum.
- Some heterozygous females may have clinical expression of the disease due to **asymmetric X-inactivation**. Aka '**manifesting heterozygotes**'. Many carrier females have adult-onset cardiomyopathy